

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau
Practical – X

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Class: TYIT	Batch: 01
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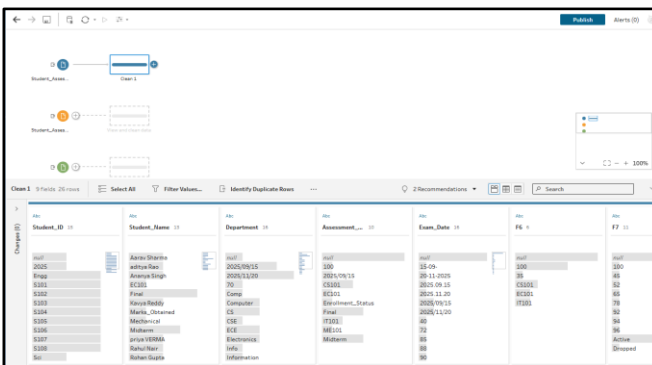
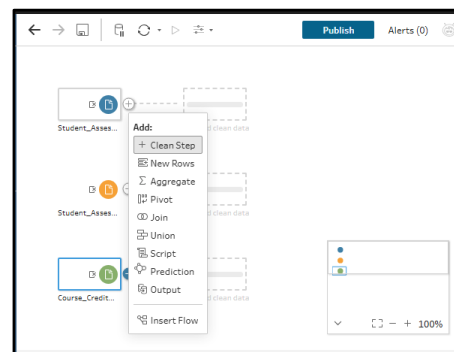
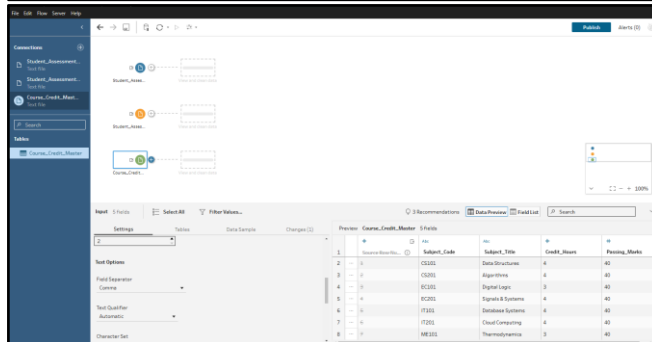
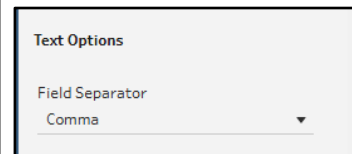
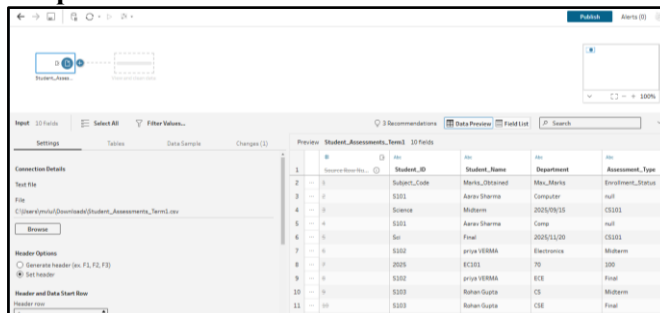
Data Preparation and Transformation using Tableau Prep

a) Connect and load data into Tableau Prep

Steps:

1. Open Tableau Prep Builder.
2. Click Connect to Data.
3. Select Text File and open the required **Student_Assessment** file.
4. In Text Options, select Comma as the field separator.
5. Add the required table to the Flow Pane to create the Input Step.
6. Check the data preview and verify fields such as Student_ID, Student_Name, Department, Assessment_Date, Score, and Max_Marks.
7. Keep the data ready for the next cleaning step.

Output:-



b) Clean and profile data

Steps:

1. Select the Input Step.
2. Click the + icon and select Clean Step.

3. Review the data profile and check the values of all fields.
4. Identify null, incorrect, duplicate, or inconsistent values.
5. Open the field menu and check/change the Data Type where required.
6. Change **Assessment_Date** to the Date data type.
7. Review the cleaned data and verify that the fields are in the correct format.

Output:-

The screenshots illustrate the process of cleaning and standardizing data in Alteryx. The data table shows columns for Student, Department, Assessment, and Exam_Date. The 'Assessment_Date' field is being changed from its default type to 'Date'. The 'Data Type' menu is shown with 'String' and 'Date' options. The 'Clean' menu is also shown with options like 'Trim Spaces' and 'Remove Extra Spaces'.

c) Group and standardize values

Steps:

1. Open the Clean Step.
2. Select the Department field.
3. Click the field's three-dot menu.
4. Select Group Values → Manual Selection.
5. Select similar department values, such as Computer/Computer Science, Electronics & Communication, and Information Technology/IT.
6. Combine the selected values into one standardized group.
7. Click Done and verify the standardized Department values.

Output:-

The screenshots illustrate the initial data cleaning and grouping steps in Tableau Prep. The top view shows a workflow with two 'Clean' steps applied to 'Student_Asses...' datasets. The middle row displays three 'Group Values by Manual Selection' dialog boxes, where specific values are being grouped for 'Department', 'Assessment...', and 'Exam_Date'. The bottom row shows a 'Department' field card and another 'Group Values by Manual Selection' dialog for 'Information Technology'.

d) Merge mismatched fields**Steps:**

1. Add the required **Student_Assessment** datasets to the Tableau Prep flow.
2. Create a **Clean** Step for each dataset.
3. Connect the **Clean** Steps of the two datasets.
4. Create **Union 1** to combine the datasets.
5. Check the fields in the **Union** result, such as **Table Names**, **F8**, **F9**, **Student_ID**, and **Student_Name**.
6. Check whether the fields from both datasets are properly combined.
7. Verify the final **Union** result before moving to the next step.

Output:-

The screenshots show the final steps of the data preparation process. The left view displays a flow diagram with 'Student_Asses...' datasets, 'Clean 2' steps, and a 'Join' step. The right view shows the 'Union Results' pane, which lists the columns in the union: 'Table Names', 'Student_ID', 'Student_Name', 'Department', and 'Assessment...'.

The top section shows two screenshots of the Alteryx interface. The first screenshot displays the 'Union Results' window for 'Union 1', showing 10 fields and 46 rows. It lists 'Table Names' (Clean 1, Clean 2), 'FB' (Field 3), 'F9' (Field 2), 'Student_ID' (18), and 'Student_Name' (18). The second screenshot shows the 'Clean 1' window, which is a 'Clean Step' with 'New Rows' selected. It shows the same fields as the Union Results window.

The bottom section shows two more screenshots. The first screenshot shows the 'Clean 2' window, which is a 'Clean Step' with 'New Rows' selected. It shows the same fields as the Union Results window. The second screenshot shows the 'Clean 3' window, which is a 'Clean Step' with 'New Rows' selected. It shows the same fields as the Union Results window.

e) Perform aggregations in Prep

Steps:

1. Select Clean 3 after the Union Step.
2. Click the + icon.
3. Select Aggregate.
4. Drag Department and Std_ID into Grouped Fields.
5. Drag Score into Aggregated Fields.
6. Set **Score** to Average (AVG).
7. Set **Full_Name** to Count (CNT) and **Max_Marks** to Sum (SUM) as shown in the screenshots.
8. Check the aggregated results in the profile pane.

Output:-

The top screenshot shows the 'Clean 3' window, which is a 'Clean Step' with 'New Rows' selected. It shows the same fields as the Union Results window. The bottom screenshot shows the 'Aggregate' window, which is an 'Aggregate Step' with 'New Rows' selected. It shows the same fields as the Union Results window, but with 'Score' set to Average (AVG), 'Full_Name' set to Count (CNT), and 'Max_Marks' set to Sum (SUM).

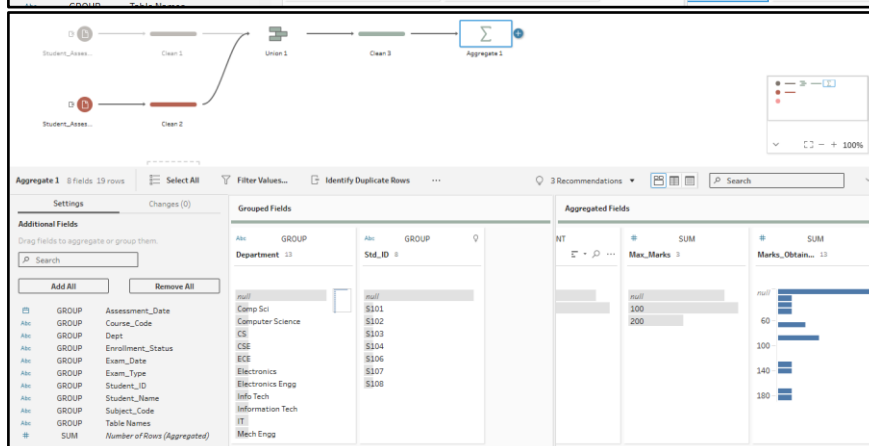
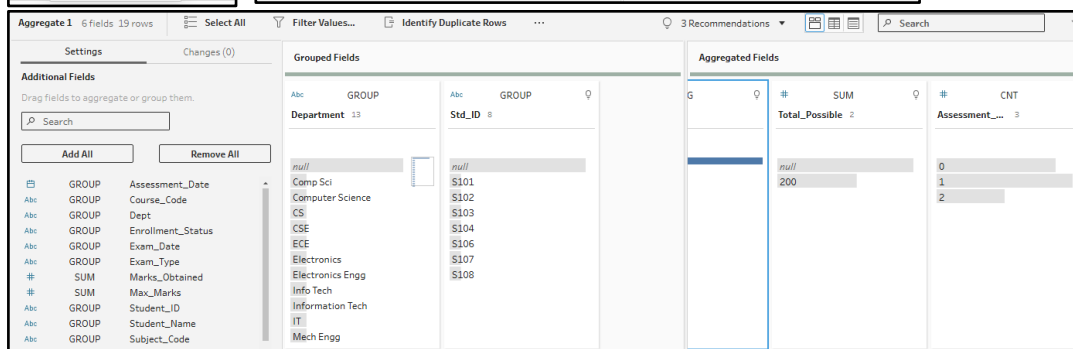
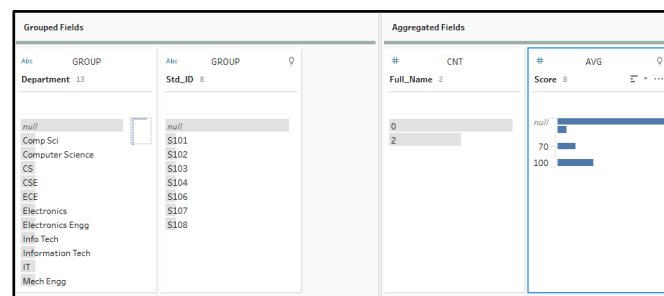
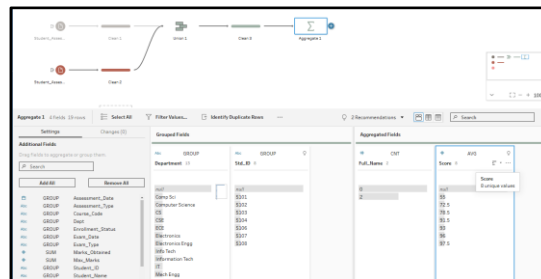
The screenshot shows the Power Query Editor interface. The data flow is as follows:

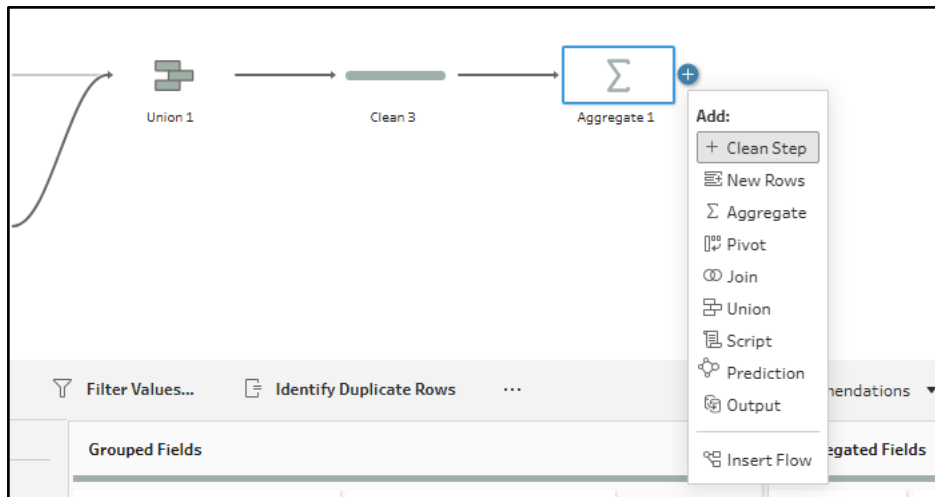
- Two source tables, both named "Student_Asses...", are loaded.
- The first source is transformed by "Clean 1" (blue bar).
- The second source is transformed by "Clean 2" (orange bar).
- The outputs of "Clean 1" and "Clean 2" are combined using a "Union" operation (purple icon).
- The result of the union is transformed by "Clean 3" (blue bar).

A context menu is open over the "Clean 3" step. The menu options are:

- Add
- + Clean Step
- New Rows
- Aggregate
- Pivot
- Join
- Union
- Script
- Prediction
- Output
- Insert Flow

At the bottom of the editor, the ribbon shows "11 fields 46 rows" and buttons for "Select All", "Filter Values...", and "Identify Duplicate Rows". The table preview at the bottom shows columns with headers "Alt" and "Abc".





Department	Std_ID	Full_Name	Score	Total_Possible	Assessment_...	Max_Mark
Comp Sci	S101				0	
Computer Science	S102				1	100
CS	S103				2	200
CSE	S104					
ECE	S106					
Electronics	S107					
Electronics Engg	S108					
Info Tech						
Information Tech						
IT						
Mech Engg						

f) Apply filters and transformations.

Filter

Steps:

1. Select Clean 4.
2. Select the Student_Name field.
3. Open the field menu.
4. Select Filter → Selected Values.
5. Select the required value; in the screenshot, 2 is selected.
6. Click Done.
7. Verify the filtered result.

Formula 1 – Term Percentage

Steps:

1. Open the field menu and select Create Calculated Field.
2. Enter the Field Name as Term_percentage.
3. Enter the following formula: $\text{ROUND}([\text{Score}] / [\text{Total_Possible}] * 100, 2)$
4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the newly created Term_percentage field.

Formula 2 – Letter Grade

Steps:

1. Select Create Calculated Field.
2. Enter the Field Name as Letter_Grade.
3. Enter the formula:
 IF [Term_percentage] >= 85 THEN "A (Distinction)"
 ELSEIF [Term_percentage] >= 75 THEN "B (First Class)"
 ELSEIF [Term_percentage] >= 60 THEN "C (Second Class)"
 ELSEIF [Term_percentage] >= 40 THEN "D (Pass Class)"

ELSE "F (Fail / Re-appear)"

END

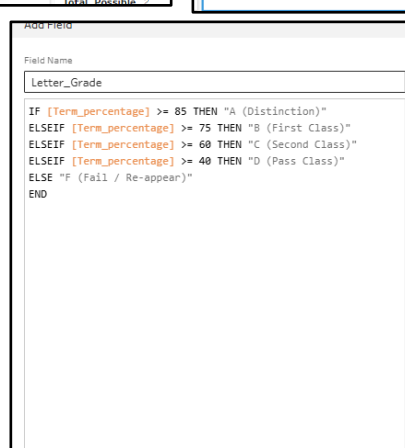
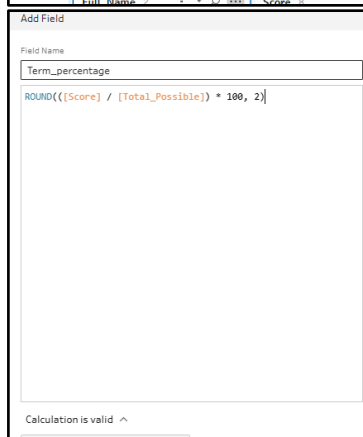
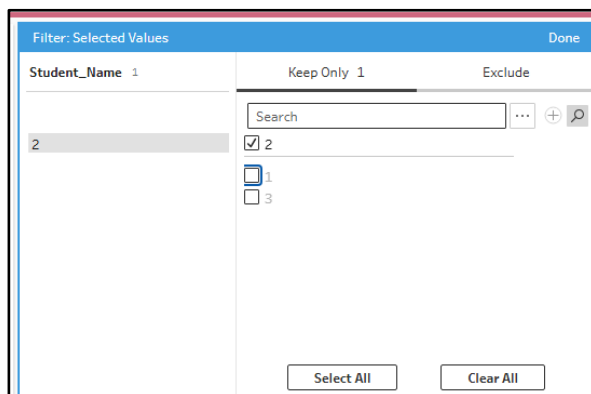
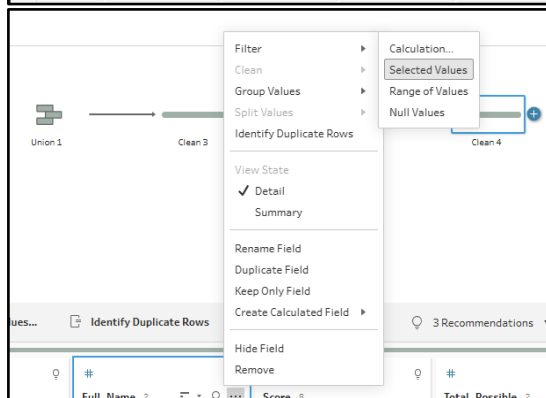
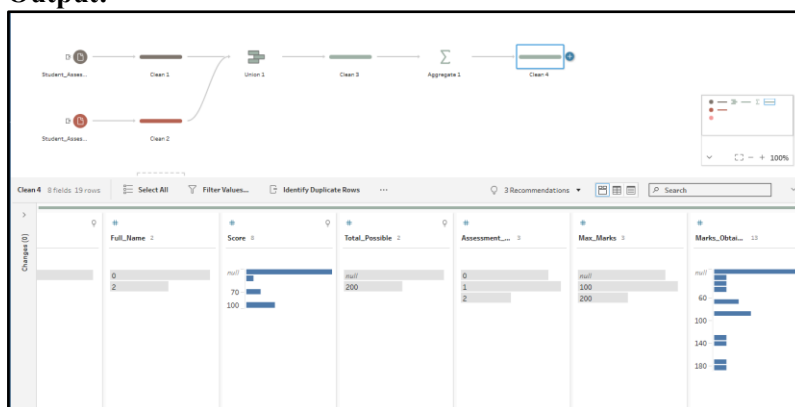
4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the **Letter_Grade** field.

Formula 3 – Academic Standing

Steps:

1. Select Create Calculated Field.
2. Enter the Field Name as **Academic_Standing**.
3. Enter the formula:
IF [Term_percentage] >= 50 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the **Academic_Standing** field.

Output:-



Add Field

Field Name

Academic_Standing

```
IF [Term_percentage] >= 60 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
```

g) Create and automate data preparation flows.

Steps:

1. Review the complete flow: Input → Clean → Union → Clean → Aggregate → Clean.
2. Select the final Clean 4 step.
3. Click the + icon on the right side of Clean 4.
4. Select Output from the menu.
5. Set Save Output Type to File.
6. Enter the required Output Name and Location.
7. Check the output preview containing fields such as Letter_Grade, Term_percentage, Department, Std_ID, Full_Name, Score, and Total_Possible.
8. Click Run Flow to execute the complete data preparation flow.
9. Verify the generated output dataset.

Output:-

Output 11 fields

Save output to: File

Save to Output.hyper

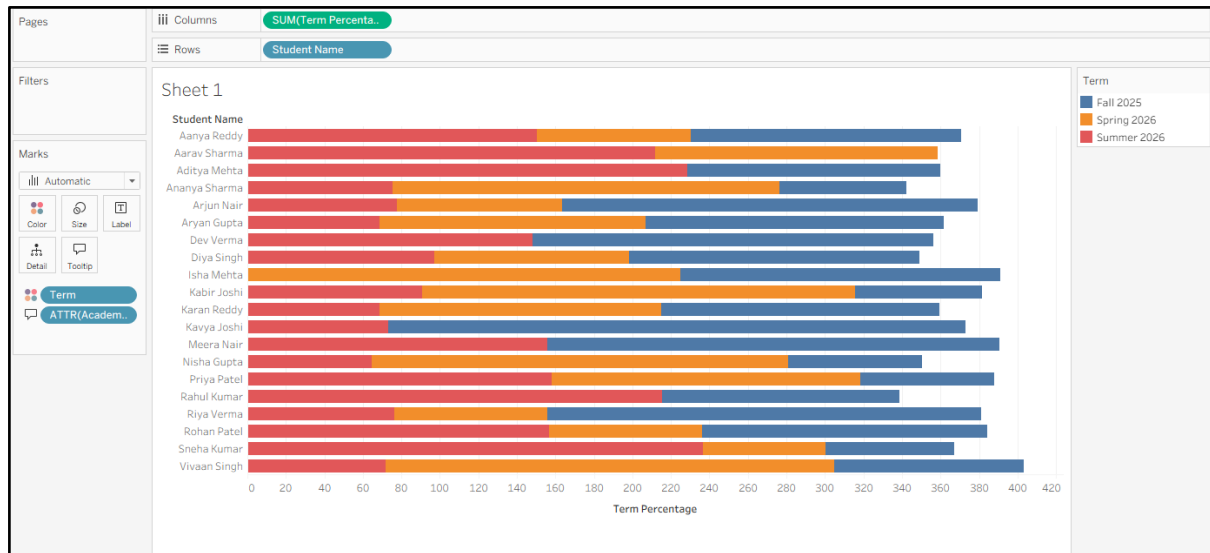
Letter_Grade	Term_percentage	Department	Std_ID	Full_Name	Score	Total_Possible
D (Pass Class)	48	null	S104	2	96	200
F (Fail / Re-appear)	27.5	null	S103	2	55	200
D (Pass Class)	48.75	null	S108	2	97.5	200
F (Fail / Re-appear)	36.25	null	S107	2	72.5	200
D (Pass Class)	45.75	null	S101	2	91.5	200
D (Pass Class)	46.5	null	S106	2	93	200
F (Fail / Re-appear)	39.25	null	S102	2	78.5	200

Letter_Grade	Term_percentage	Department	Std_ID	Full_Name	Score
D (Pass Class)	48	null	S100	2	96
F (Fail / Re-appear)	27.5	null	S100	2	55
D (Pass Class)	48.75	null	S100	2	97.5
F (Fail / Re-appear)	36.25	null	S107	2	72.5
D (Pass Class)	45.75	null	S101	2	91.5
D (Pass Class)	46.5	null	S106	2	93
F (Fail / Re-appear)	39.25	null	S102	2	78.5

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D (Pass Class)	48	null	S100	2	96
F (Fail / Re-appear)	27.5	null	S100	2	55
D (Pass Class)	48.75	null	S100	2	97.5
F (Fail / Re-appear)	36.25	null	S107	2	72.5
D (Pass Class)	45.75	null	S101	2	91.5
D (Pass Class)	46.5	null	S106	2	93
F (Fail / Re-appear)	39.25	null	S102	2	78.5

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D (Pass Class)	48.75	null
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D (Pass Class)	45.75	null
D (Pass Class)	46.5	null
F (Fail / Re-appear)	39.25	null



Overall Conclusion

The complete data preparation process was performed successfully in Tableau Prep. The data was loaded, cleaned, standardized, combined, aggregated, filtered, and transformed using calculated fields. Finally, the prepared data was sent to an Output Step and the complete flow was executed successfully.